

Kali Linux Essentials & Fundamentals Guide

File System Architecture, Core Commands, Permissions & Package Management

1. Linux File System Architecture Overview

Unlike Windows, which uses drive letters (C:, D:), Linux organizes all directories and files into a single, unified tree structure originating from the root directory ('/'). Below is an overview of the key directories covered in this lab:

Directory	Purpose	Description & Key Contents
/	Root Directory	The top-level root of the entire file system hierarchy. All files and directories stem from here.
/home	User Directories	Contains personal workspace directories for standard system users (e.g., /home/kali).
/etc	Configuration Files	Stores system-wide configuration files and administrative shell scripts (e.g., /etc/passwd, /etc/network).
/var	Variable Data	Holds dynamic, frequently changing data such as system logs (/var/log), mail, and database spools.

2. Command Cheat-Sheet & Practice Report

This section details 20 foundational Linux commands practiced in the Kali Linux terminal environment, complete with command purpose and simulated terminal outputs.

2.1 Command: pwd — Print Working Directory

Purpose: Displays the full absolute path of the current directory.

```
(kali@kali)-[~]
$ pwd
/home/kali
```

2.2 Command: ls — List Directory Contents

Purpose: Lists files and subdirectories within the current path.

```
(kali㉿kali)-[~]  
└─$ ls  
Desktop  Documents  Downloads  Music  Pictures  Public  Videos
```

2.3 Command: cd — Change Directory

Purpose: Navigates between directories in the system hierarchy.

```
(kali㉿kali)-[~]  
└─$ cd /var/log && pwd  
/var/log
```

2.4 Command: cat — Concatenate & Display

Purpose: Reads and prints the full contents of a file to the screen.

```
(kali㉿kali)-[~]  
└─$ cat /etc/hostname  
kali
```

2.5 Command: mkdir — Make Directory

Purpose: Creates one or more new directories.

```
(kali㉿kali)-[~]  
└─$ mkdir -p project_files && ls -d project_files  
project_files
```

2.6 Command: touch — Create Empty File

Purpose: Updates timestamp or creates a new empty file if it doesn't exist.

```
(kali㉿kali)-[~]  
└─$ touch testfile.txt && ls testfile.txt  
testfile.txt
```

2.7 Command: rm — Remove Files/Directories

Purpose: Deletes files or directories recursively.

```
(kali㉿kali)-[~]  
└─$ rm testfile.txt && ls testfile.txt  
ls: cannot access 'testfile.txt': No such file or directory
```

2.8 Command: cp — Copy Files/Folders

Purpose: Copies files or directories from source to target destination.

```
(kali㉿kali)-[~]  
└─$ cp /etc/issue ./issue_copy.txt && cat issue_copy.txt  
Kali GNU/Linux Rolling \n \l
```

2.9 Command: mv — Move/Rename

Purpose: Moves files to another location or renames them.

```
(kali㉿kali)-[~]  
└─$ mv issue_copy.txt renamed_issue.txt && ls renamed_issue.txt  
renamed_issue.txt
```

2.10 Command: echo — Print Text / Output String

Purpose: Prints text strings or redirects them into files.

```
(kali㉿kali)-[~]  
└─$ echo 'Hello Kali Security' > greeting.txt && cat greeting.txt  
Hello Kali Security
```

2.11 Command: head — View First Lines

Purpose: Displays the first 10 lines of a target file.

```
(kali㉿kali)-[~]  
└─$ head -n 3 /etc/passwd  
root:x:0:0:root:/root:/bin/bash  
daemon:x:1:1:daemon:/usr/sbin:/bin/sh  
bin:x:2:2:bin:/bin:/bin/sh
```

2.12 Command: tail — View Last Lines

Purpose: Outputs the final lines of a file; supports live tracking (-f).

```
(kali㉿kali)-[~]  
└─$ tail -n 3 /etc/passwd  
statd:x:114:118:./var/lib/nfs:/usr/sbin/nproc  
sshd:x:115:65534:./run/sshd:/usr/sbin/nologin  
kali:x:1000:1000:Kali,,,:/home/kali:/bin/zsh
```

2.13 Command: grep — Search Text Patterns

Purpose: Searches files or output matching specific regex strings.

```
(kali㉿kali)-[~]  
└─$ grep 'kali' /etc/passwd  
kali:x:1000:1000:Kali,,,:/home/kali:/bin/zsh
```

2.14 Command: chmod — Change File Modes/Permissions

Purpose: Modifies read, write, and execute permissions of files.

```
(kali㉿kali)-[~]  
└─$ chmod 755 project_script.sh && ls -l project_script.sh  
-rwxr-xr-x 1 kali kali 45 Aug 26 09:00 project_script.sh
```

2.15 Command: chown — Change Ownership

Purpose: Changes user and group ownership of files.

```
(kali㉿kali)-[~]  
└─$ sudo chown root:root project_script.sh && ls -l project_script.sh  
-rwxr-xr-x 1 root root 45 Aug 26 09:00 project_script.sh
```

2.16 Command: sudo — Superuser Do

Purpose: Executes commands with administrative (root) privileges.

```
(kali㉿kali)-[~]  
└─$ sudo whoami  
root
```

2.17 Command: uname — System Information

Purpose: Prints kernel name, version, and architecture.

```
(kali㉿kali)-[~]  
└─$ uname -a  
Linux kali 6.8.11-kali1-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.8.11-1 (2024-05-29) x86_64  
GNU/Linux
```

2.18 Command: df — Disk Space Usage

Purpose: Reports filesystem disk space consumption.

```
(kali㉿kali)-[~]  
$ df -h /  
Filesystem      Size  Used Avail Use% Mounted on  
/dev/sda1       80G   18G   58G   24% /
```

2.19 Command: free — Memory Usage

Purpose: Displays system total, used, and available RAM.

```
(kali㉿kali)-[~]  
$ free -m  
              total        used        free      shared  buff/cache   available  
Mem:          7942         1420         4850          120        1672        6100
```

2.20 Command: whoami — Current User ID

Purpose: Prints the effective user name associated with the current shell.

```
(kali㉿kali)-[~]  
$ whoami  
kali
```

3. Hands-On Practical Tasks

Task 1: Folder & File Creation, Writing, and Reading Text

In this exercise, a dedicated directory is created, a text file is initialized with written content, and its contents are read back using cat.

```
(kali㉿kali)-[~]  
$ mkdir -p ~/kali_practice && cd ~/kali_practice  
echo 'Kali Linux File Handling Lab' > lab_notes.txt  
cat lab_notes.txt  
Kali Linux File Handling Lab
```

Task 2: File Permission Management with chmod & Ownership with chown

Permissions in Linux are calculated using octal values (4=Read, 2=Write, 1=Execute). Standard permission masks include:

- 755 (rwxr-xr-x): Owner has full permissions (Read, Write, Execute); Group and Others have Read and Execute.
- 644 (rw-r--r--): Owner can Read and Write; Group and Others have Read-only access.

```
(kali㉿kali)-[~]
└─$ # Change permission to 755 (Owner: rwx, Group: r-x, Others: r-x)
chmod 755 lab_notes.txt
ls -l lab_notes.txt

# Change file ownership to root using sudo
sudo chown root:root lab_notes.txt
ls -l lab_notes.txt
-rwxr-xr-x 1 kali kali 28 Aug 26 09:00 lab_notes.txt
[sudo] password for kali:
-rwxr-xr-x 1 root root 28 Aug 26 09:00 lab_notes.txt
```

Task 3: Package Management with Advanced Package Tool (apt)

The Advanced Package Tool (apt) manages debian-based packages. Installing tools requires root privileges using sudo.

```
(kali㉿kali)-[~]
└─$ # Update package repositories and install nmap
sudo apt update && sudo apt install -y nmap
Hit:1 http://http.kali.org/kali kali-rolling InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
nmap is already the newest version (7.94SHA1+SVN-6kali1).
0 upgraded, 0 newly installed, 0 to remove.
```

```
(kali㉿kali)-[~]
└─$ # Verify nmap installation
nmap --version
Nmap version 7.94SVN ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: nmap-liblua-5.4.6 openssl-3.0.13 libssh2-1.11.0 libz-1.3.1
```

```
neha@kali: ~/internship
Session Actions Edit View Help
neha@kali: ~/internship neha@kali: ~/Downloads
(neha@kali)-[~]
$ pwd
/home/neha
(neha@kali)-[~]
$ ls
Desktop Documents Downloads intership Music Pictures Projects Public scan.txt simplectf Templates Videos
(neha@kali)-[~]
$ mkdir intership
(neha@kali)-[~]
$ cd intership
(neha@kali)-[~/internship]
$
```

```
neha@kali: ~/internship
Session Actions Edit View Help
neha@kali: ~/internship neha@kali: ~/Downloads
(neha@kali)-[~/internship]
$ touch text.txt
(neha@kali)-[~/internship]
$ ls
text.file text.txt
(neha@kali)-[~/internship]
$ rm text.txt
(neha@kali)-[~/internship]
$
```

```
Session Actions Edit View Help
neha@kali: ~/internship | neha@kali: ~/Downloads |
(neha@kali)-[~/internship]
$ ls
text.file
(neha@kali)-[~/internship]
$ nano text.file
(neha@kali)-[~/internship]
$ cat text.file
Hello,My name is NEHA.
(neha@kali)-[~/internship]
$ ls -l
total 4
-rw-rw-r-- 1 neha neha 24 Aug 27 18:17 text.file
(neha@kali)-[~/internship]
$ chmod 755 text.file
(neha@kali)-[~/internship]
$ ls -l
total 4
-rwxr-xr-x 1 neha neha 24 Aug 27 18:17 text.file
(neha@kali)-[~/internship]
$
```

```
Session Actions Edit View Help
neha@kali: ~/internship | neha@kali: ~/Downloads |
(neha@kali)-[~/internship]
$ sudo apt update && sudo apt install -y nmap
[sudo] password for neha:
Get:1 http://mirror.yer.az/kali kali-rolling InRelease [34.0 kB]
Get:2 http://mirror.yer.az/kali kali-rolling/main amd64 Packages [21.4 MB]
Get:3 http://mirror.yer.az/kali kali-rolling/main amd64 Contents (deb) [53.3 MB]
Get:4 http://mirror.yer.az/kali kali-rolling/contrib amd64 Packages [115 kB]
Get:5 http://mirror.yer.az/kali kali-rolling/contrib amd64 Contents (deb) [265 kB]
Get:6 http://mirror.yer.az/kali kali-rolling/non-free amd64 Packages [183 kB]
Get:7 http://mirror.yer.az/kali kali-rolling/non-free amd64 Contents (deb) [915 kB]
Fetched 76.2 MB in 52s (1,473 kB/s)
1134 packages can be upgraded. Run 'apt list --upgradable' to see them.
nmap is already the newest version (7.99+dfsg-1kali1).
nmap set to manually installed.
Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 1134
```

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Linux Commands Reference

<https://www.geeksforgeeks.org/linux-unix/basic-linux-commands/>

<https://tryhackme.com/module/linux-fundamentals>